24. DAY 1 REVISION: THE NODAL FLOW

DURATION : 03:43 STUDY SHEET

COURSE SUMMARY

This session addresses the origin of internal asymmetry through Hensen's node and nodal flow. Rotational cilia create a movement that allows the displacement of molecules and vesicles, thus generating morphogens that influence the asymmetrical development of organs, such as the heart and liver. The genetic switches involved, such as Sonic Hedgehog and fibroblast growth factors (FGF), play a key role in the organisation of internal structures.

The embryonic axis is presented as an organising centre, with an emphasis on the importance of the intra-embryonic mesoderm. The impact of these concepts on osteopathic practice is also highlighted, by showing how understanding these processes is crucial for patient care, thanks to an appropriate mental image of these internal dynamics.

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THE ORIGIN OF INTERNAL ASYMMETRY: HENSEN'S NODE

At the origin of **internal asymmetry**, at **Hensen's node**, small **cilia** are observed. These cilia do not perform a simple movement, but a **rotational movement**.

They accelerate when they are higher relative to their base and bring small **molecules**. These molecules come from the "roof" (from inside our body) towards the inside of the node.

It's a big "mess" where cilia are present. Above, small **vesicles** fall and concentrate towards a specific side.

VESICLES AND MORPHOGENS

These small vesicles collide due to "excess" and release a large number of **morphogens** onto the ground below. There are **activation genes**, which amplify the phenomena, and **inhibition genes**.

It is this genetic mechanism that will create the structures of **asymmetry**. Later, this will determine that the **heart** will develop on the left, the **liver** more to the right, and that the internal organs will be asymmetrical.

We do not have perfect **symmetry**, but rather **functional asymmetry** which will give us long-term movements. All of this is organised down to the **cerebral connections**.

THE STARTING POINT OF ASYMMETRY AND NODAL FLOW

Nodal flow is the starting point of asymmetry. It is this small movement of the cilia that moves the **nodal fluid** and concentrates the vesicles more on one side than the other.

The vesicles are of **epiblastic** origin, coming from the roof of the epiblast. It is here that the entire cascade, left and right, will occur, with this concentration towards the right.

This will trigger cascades involving important genes such as **Sonic Hedgehog** and genes from the **FGF** (Fibroblast Growth Factors) family. These genes will be involved up to Track 2 and will be important later for the creation of new structures, new **proteins**, and for the organisation of asymmetry.

THE AXIS AS A CENTRE OF ORGANISATION

This organisation is established very early. This **axis** is a true centre of organisation.

The epithelial cells called "**bottle cells**" and the **mesenchymal** cells will form the **mesoderm**. There is **extra-embryonic** and **intra-embryonic** mesoderm, but it is mainly the intra-embryonic mesoderm that interests us here.

It essentially originates from the epiblastic epithelium. This is why, initially, it is first **electricity** that induces and organises the fluid, and then the fluid that organises matter.

THE IMPORTANCE IN OSTEOPATHY

This is crucial in our **osteopathic** thinking, in our work. We work with our **mental image**.

What is our mental representation of these processes? The body uses it, whether we are aware of it or not.